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Research Program/Project/Study Title: Level of Disaster Preparedness and Resiliency of the Highly Devastated Communities of Roxas, City, Capiz	
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Program or Project has component (if no proceed to **2**)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☒ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center							
(4) Site of implementation									
		Municipality/City:	Roxas City						
		Province:	Capiz						
		Region:	VI						
(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☒ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☐ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☒ Basic	☐ Pilot Testing	☐ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
Research:	Development:								
☒ Basic	☐ Pilot Testing								
☐ Applied	☐ Tech. Promotion/Commercialization								
(7) Sector/Commodity		(8) Discipline/Field of Specialization Criminology							

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(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

Disaster preparedness is essential for homes, businesses, and communities, but many people are still ill-prepared. Recent disasters have shown how important individual accountability, local cooperation, and continuity plans are for ensuring that significant events can be responded to and recovered from.

The Philippines is the country with the second-highest risk of natural disasters (UNU-EHS and ADW-2014 as cited by Andriesse 2018). It is susceptible to humanitarian disasters because to its rising population, island-based environment, and inadequate infrastructure (Lum and Margesson 2014). It is the nation most frequently hit by tropical cyclones in the globe, with an average of 20 per year, of which 6 are deemed destructive. It is located in the north-west Pacific Ocean. It is also located immediately below the Inter-tropical Convergence Zone, and the northeast and southwest monsoons. These cause fatalities and damage to buildings and homes worth billions of pesos (Pealba et al. 2012; Lee and Vink 2015; Cas 2016; Enteria 2016). Due to its location along the Ring of Fire in the Pacific Ocean, this nation is susceptible to earthquakes, tsunamis, and volcanic eruptions (Pe Symaco 2013).

Storm surges from Haiyan (Yolanda) in 2013 reached heights of 6-7 m. The coastal settlements were severely damaged by the strongest tropical cyclone and bloodiest calamity in Philippine contemporary history (Center for Research on the Epidemiology of Disasters 2014 as cited by Dalisay and De Guzman 2016; Nakamura et al. 2016). The National DRRM Council of the nation reported approximately 6300 fatalities, 1061 missing persons, and 28,689 wounded. 47,000 shelters were almost damaged, while over 12,000 were completely demolished (Commission on Audit [COA] Report 2014 as cited by Dalisay and De Guzman 2016).

In April 2022, Typhoon Agaton struck the Island of Panay with 145,713 effected persons including 67,184 who evacuated from their homes and 38,598 persons in the evacuation centers. In the provincial level, the Province of Capiz was put under the state of calamity after the onslaught of the tropical depression Agaton with 19 confirmed deaths and an estimated damage of Php 176 million in the agriculture sector, Php 500 million in the fishery, and infrastructure damage of Php 59 million with many areas flooded for three days. pna.gov.ph/articles/1172524.

Recently, on October 29, 2022 another Severe Tropical Storm named “Paeng” flooded the Province of Capiz after the heavy and intense rain which affected 309 barangays, 78 of which are “hard hit” areas in all 17 towns. With 6 deaths (4 of which were from drowning) including a town councilor, 317,500 displaced persons due to heavy flooding and landslides classified has put the Province to be the most affected province in Western Visayas. pna.gov.ph/articles/1187327

In the City of Roxas, the most devastated communities according to the data of the City Disaster Risk Reduction Management Office are Barangay Adlawan, Loctugan, Lanot, San Jose, Bolo, and Liong in terms of economic damage in agriculture, fisheries, livestock, business, and infrastructure due to days of flooding.

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Contrary to the initial expectations of the researchers that the coastal communities of Culasi, Baybay, Dumolog, Olotayan, Libas, Barra, and Cogon are the most vulnerable and devastated.

With that being said, cognizant with the vision of Capiz State University in line with Delivering Quality Service to All, the Criminal Justice Education commits itself along with its core functions along with research, instruction, production, and extension by providing relevant trainings on disaster preparedness, planning, response, and resiliency. The purpose of this research is to determine the level of disaster preparedness and resiliency of the identified community with the highest number of casualty and economic damage in terms of agriculture, fisheries, livestock, business, and infrastructure. The data of this research will be used as a baseline data to determine the community to be adopted and trained by the Criminal Justice Education in their extension program.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)

The data on the number of casualties and economic damage sustained by the 47 barangays of Roxas City will be requested and gathered from the City Disaster Risk Reduction Management Office at Bilbao Street, Roxas City. Only five communities with the highest number of human casualty (if any) and highest economic damage in terms of agricultural, fisheries, livestock, business, and infrastructure will be chosen.

The questionnaires will be distributed to the residents of the chosen communities. The result of this research will be used as a baseline data to determine the community to be adopted and trained by the Criminal Justice Education in their extension program.

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

This study will be conducted to determine the level of Disaster Preparedness and Resiliency of the Coastal Communities in Roxas City. Specifically, this sought to answer the following questions?

1. What is the socio-demographic profile of the respondents?
2. What is the level of Disaster of Preparedness and Resiliency of the five most disaster devastated community in Roxas City?
3. Is there a significant difference in the level of Disaster Preparedness and Resiliency of the five most disaster devastated community in Roxas City when grouped according to location?

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

In the study conducted by Eun Lee (2013), majority of deaths recorded during a meteorological disaster while occupation with the most number of casualties are in agriculture, forestry, and fishery. Lastly, the most common cause of deaths during a disaster is drowning.

In the study of Dariagan (2021), Despite of the fact that the Philippines is a disaster prone country

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because of its location in the pacific and the number of natural disaster that visit it every year, and despite of the vulnerability of most of the 92 local government units in Panay Island they were still found to be partially prepared with the exception of coastal, middle – earning, most populated LGUs to have higher level of Preparedness.

The over 14 million seriously-affected Filipinos were inhabitants of the Visayas zone (Hall and Ashcroft 2014; O’Connell et al. 2017) in central Philippines particularly Eastern, Central and Western Visayas regions. The scourged provinces in Western Visayas were Capiz, northern part of Iloilo, Aklan, and northern portion of Antique all located in Panay Island.

Every disaster has brought with it a number of economic constraints that affect the nation's economic growth and efforts to reduce poverty (Arnold 2006; Bui et al. 2014; Rodriguez-Oreggia et al. 2013), but this could be avoided when there is good management in putting the planned effective measures into action (Quarantelli 2002). Effective planning and execution are complemented by investments in personnel and machinery.

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Descriptive correlational research design will be used to facilitate the data needed for this research. Descriptive research design is a type of research design that aims to obtain information to systematically describe a phenomenon, situation, or population. While correlational research design investigates relationships between variables without the researcher controlling or manipulating any of them.

Descriptive research design will be used to gather data regarding the socio-demographic profile of the selected communities and their level of disaster preparedness and resiliency. While correlational research design will be used to determine if there is significant difference in the level of disaster preparedness and resiliency of the five most disaster devastated communities of Roxas City, Capiz when grouped according to location (Barangay).

The data on the number of casualties and economic damage sustained by the 47 barangays of Roxas City will be requested and gathered from the City Disaster Risk Reduction Management Office at Bilbao Street, Roxas City. Only five communities with the highest number of human casualty (if any) and highest economic damage in terms of agricultural, fisheries, livestock, business, and infrastructure will be chosen.

Information of the disaster preparedness and resiliency will be gathered using a survey questionnaire. The sample size of the respondents will be determined using the proportional allocation formula by Pagoso (1985):

$$n_i = \frac{n_T N_i}{N_T}$$

n_i = sample size per category

n_T = population per category

N_i = total sample size

N_T = total population

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The survey questionnaire to be used in gathering data will be composed of three (3) parts. Part I t will be used to gather the socio-demographic profile of the respondents. The second part will be used to gather the data on the level of disaster preparedness of the community. The third part of the questionnaire will gather the level of disaster resiliency of the selected communities in Roxas City.

The survey questionnaire will be administered to the respondents of the selected coastal community by the researchers. The data will be analyzed using frequency count, mean and analysis of variance.

Scoring Guide on the level of Disaster Preparedness and Resiliency of Coastal Communities of Roxas City, Capiz

Score	Response Category
5	very well prepared
4	well prepared
3	prepared
2	quite prepared
1	Not prepared

To interpret the data on the level of Disaster Preparedness and Resiliency of Coastal Communities of Roxas City, Capiz, the following mean interval and verbal interpretation were used as basis:

Mean Interval	Verbal Interpretation
4.20 -5.0	very well prepared
3.40 4.19	well prepared
2.60 – 3.39	prepared
1.80 – 2.59	quite prepared
1.0 – 1.79	Not prepared

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.

	Year One										
	Aug	Sept	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
1. Field work/Survey											
2. Analysis of data gathered from observation											

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

Product:

- Sustainable disaster preparedness plan
- Data on the level of disaster preparedness and resiliency of selected coastal barangays communities of Roxas City

People

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- Capability building (highly devastated communities)

Partnership: MOA

Potential Outcomes:

Potential Impacts

1. Enhance the awareness of disaster prone communities on disaster related preparation.
2. Research publication

(15) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

Through extension activities

(16) Targets Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

Local Government Units and Residents of the highly devastated communities of Roxas City during disasters.

(17) Personnel Requirement

Name	Role (Program,Project, Study Leader)	Field of Specialization
Mr. Jason B. Capundan	Study Leader	Criminology
Dr. Sebastian C. Caduco Jr.	Project Member	Criminology
Dr. Verna R. Belarmino	Project Member	Criminology
Mr. Carlo M. Salido	Project Member	Criminology
Dr. Jocelyn S. Legaspi	Project Member	Fisheries
Prof. Jocel C. Sitjar	Project Member	Criminology
Mr. Allan F. Buhayan	Project Member	Criminology
Mr. Alfonso B. Clavero Jr	Project Member	Criminology
Mr. Den Mark G. Orquia	Project Member	Criminology

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

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(18) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (Php)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)		
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		
- Transportation allowance	Travel from Capsu to the selected coastal barangays	Php 3,000.00
- Meal allowance during data gathering		
- Professional fees (laboratory analysis and other related fees)	Statistician	Php 5,000.00
- Miscellaneous (for incidental expenses)		Php 2,000.00
- Printing	Coupon bond and Ink	Php 3,000.00
C. CAPITAL OUTLAY		
Total		Php 13,000.00

(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

Cavallo E, Galiani S, Noy I, Pantano J (2013) Catastrophic natural disasters and economic growth. Rev Econ Stat 95:1549–1561. <https://doi.org/10.1162/REST-a-00413>

Commission on Audit (2014) Disaster management practices in the Philippines: an assessment. COA Report. https://www.coa.gov.ph/disaster_audit/doc/National.pdf. Accessed 18 Jan 2015

Dalisay SN, De Guzman MT (2016) Risk and culture: the case of typhoon Haiyan in the Philippines. Dis Prev Manag 25:701–714. <https://doi.org/10.1108/DPM-05-2016-0097/full/html>

Department of the Interior and Local Government. LGU Disaster Preparedness Journal: Checklist of Minimum Actions for Mayors. <http://downloads.caraga.dilg.gov.ph/Disaster%20Preparedness/Journal.pdf>. Accessed 9 Mar 2015

Hall M, Ashcroft A (2014) Connecting and communicating after typhoon Haiyan. Forced Migr Rev 45:95–96

Hashemipour M, Stuban SME, Dever JR (2017) A community-based disaster coordination framework for effective disaster preparedness and response. Aust J Emerg Manag 32:41–46

Lagmay AMF, Agaton RP, Bahala MAC, Briones JBLT, Cabacaba KMC, Caro CVC, Mungcal MTF (2015) Devastating storm surges of Typhoon Haiyan. Int J Disaster Risk Reduct 11:1–12. <https://doi.org/10.1016/j.ijdr.2014.10.006>

Lum T, Margesson R (2014) Typhoon Haiyan (Yolanda): U.S. and international response to Philippines disaster. Current politics and economics of south, South-eastern, and central Asia 23:209–246. <https://search.proquest.com/openview/4583721e1283e9672756f878840eec7a/1?pq->

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origsite=gscholar&cbl=2034881. Accessed 20 January 2020

Lee, K. E., Myung, H. N., Na, W., & Jang, J. Y. (2013). Socio-demographic characteristics and leading causes of death among the casualties of meteorological events compared with all-cause deaths in Korea, 2000-2011. *Journal of preventive medicine and public health*, 46(5), 261.

(20) Capsule Curriculum Vitae (One-page brief C.V. only)

JASON B. CAPUNDAN

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Academic History

Masters Capiz State University , Dayao Satellite College, 2019, Master of Science in Criminal Justice major in Criminology

Tertiary Filamer Christian University, 2015, Bachelor of Science in Criminology

Professional Experience:

- **Instructor 1**
Capiz State University , Dayao Satellite College
2020 to date
- **Full – time Faculty**
Filamer Christian University, College of Criminal Justice Education
2016 – 2020

Licenses and Certification:

- Registered Criminologist

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Level of Disaster Preparedness of the Coastal Communities in Roxas, City, Capiz

Dear Respondents:

We are carrying out an evaluation on the level of Disaster Preparedness and Resiliency in your community. Thus, the researchers would like to solicit your answers by responding to the questions below with utmost sincerity. Your answers will be treated with complete confidentiality. Thank you!

Researchers

Part I. Socio – demographic Profile of the Respondents

Direction: Kindly provide complete information for each item by putting a check mark (/) on the space provided.

1. Name (Optional): _____
2. Age: _____
3. Gender: _____
4. Educational Attainment:
 - _____ Elementary level
 - _____ Elementary Graduate
 - _____ High School Level
 - _____ High School Graduate
 - _____ College Undergraduate
 - _____ College Graduate
 - _____ Post Graduate
 - _____ others (please specify _____)
5. Role in the family: _____ Head of the household _____ member of the household
6. Household size: _____
7. Occupation: _____
8. Sources of income: _____
- 10 Type of houses: ___ light materials ___ combination(light/concrete) ___ concrete
- 11 House/business location :
 - _____ River bank
 - _____ 50 meters from the coast/river
 - _____ 100 meters from the coast/river
 - _____ 150 meters from the coast/river
 - _____ 200 Meters from the coast/river
 - _____ 250 meters or more away from the coast/river

A. Level of disaster preparedness

Overall review rating	Equivalent rating in studies
Very well prepared	At least 90% of the respondents feel confident/prepared
Well prepared	At least 80% but fewer than 90% of respondents feel confident /prepared

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Prepared	At least 60% but fewer than 80% of respondents feel confident/prepared
Quite prepared	At least 40% but fewer than 60% of respondents feel confident/prepared
Not prepared	At least 20% but fewer than 80% of respondents feel confident/prepared

Direction. Below are statements which describes the level of disaster prepared of your community which are categorized as **Very well prepared, well prepared, prepared, quite prepared, and not prepared**. Kindly put a check mark (✓) on the space provided which corresponds to your answer.

Level of Disaster Preparedness of Coastal Communities of Roxas City, Capiz

Statement	Very well prepared (5)	Well prepared (4)	Prepared (3)	Quite prepared (2)	Not prepared (1)
a. Family Emergency Kit such First Aid Kit and Hygiene Items					
b. Over-the-counter medicines for fever, cough, and colds					
c. Means of Communication such as Telephones, Cellphones, or handheld radios					
d. Lists of emergency numbers such as MDRRMC,BFP, and PNP					
e. Battery-powered radio and flashlights					
f. Extra Batteries					
g. Evacuation plan for the family in-case of a natural or man-made disaster					
h. Food for 3 days for the entire family					
i. Water supply that is good for 3 days for the entire family					

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j. Extra Clothing					
k. Rain Coat					
l. Emergency Whistle					
m. Emergency Boots					
n. Floatation Devices					
o. Sprinkler System					
p. Fire Extinguishers					
q. Smoke Detectors					
r. Harness and ropes to strengthen the house					

B. Level of disaster resiliency

Overall review rating	Equivalent rating in studies
Always	At least 90% of respondents apply/have in disaster situations
Often	At least 80% but lower than 90% of respondents apply/have in disaster situations
Sometimes	At least 60% but lower than 80% of respondents apply/have in disaster situations
Rarely	At least 40% but lower than 60% of respondents apply/have in disaster situations
Never	At least 20% but lower than 40% of respondents apply/have in disaster situations

Direction. Below are statements which describes the level of disaster resiliency of your community which are categorized as **Always, Often, Sometimes, Rarely, and Never**. Kindly put a check mark (/) on the space provided which corresponds to your answer.

Level of Disaster Resilience of Coastal Communities of Roxas City, Capiz

Statement	Always (5)	Often (5)	Sometimes (3)	Rarely (2)	Never (1)
• Access to a wide range of damage-resistant communication devices to gather news and information regarding disasters					
• Active participation in					

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earthquake and fire drills					
Active participation and involvement in the disaster preparedness planning of the barangay council					
Activities and information disseminations regarding disaster planning and recovery has reached the community					
Community members are trained and mobilized for disaster preparedness					
Availability of well-known sites for water supply					
Identified high risk areas					
Regular conduct of structural check and maintenance					
More than 80% of the household have members skilled in house maintenance and repairs					
More than 50% of the households have members skilled in house maintenance and repairs					